

# WINTER SCHOOL SESSION ON BIOMEDICAL IMAGE PROCESSING AND ITS APPLICATIONS

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Name- Lubdhak Nairith Saha  
Arnish  
Roll- 23053432  
B.tech 3<sup>rd</sup> Year CSE



[www.linkedin.com/in/lubdhak-nairith-saha-2761182a0](https://www.linkedin.com/in/lubdhak-nairith-saha-2761182a0)

[lubdhakarnish@gmail.com](mailto:lubdhakarnish@gmail.com)

# PROBLEM STATEMENT

Deep learning models are widely used for image classification, but their performance depends on architecture, training strategy, and hyperparameters. This project studies how default, pretrained, and customized VGG16 models perform on medical and non-medical datasets.





# OBJECTIVE

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- To implement VGG16 CNN for image classification
- To compare default and pretrained models
- To analyze the effect of fine-tuning
- To evaluate accuracy and execution time



# DATASETS USED

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| Dataset                  | Type        | Classes              |
|--------------------------|-------------|----------------------|
| Chest X-ray<br>Pneumonia | Medical     | NORMAL,<br>PNEUMONIA |
| Cats vs Dogs             | Non-medical | Cat, Dog             |

# MODEL USED

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VGG16 is a deep convolutional neural network with 16 layers. It is widely used for image recognition and supports transfer learning using pretrained ImageNet weights.





# EXPERIMENTS PERFORMED

| Experiment | Description                              |
|------------|------------------------------------------|
| Model 1    | VGG16 from scratch (no pretraining)      |
| Model 2    | Pretrained VGG16 (frozen layers)         |
| Model 3    | Fine-tuned VGG16 (last layers trainable) |



# MEDICAL DATASET RESULTS (CHEST X- RAY)

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| Model                | Accuracy     | Training Time |
|----------------------|--------------|---------------|
| VGG16 (from scratch) | 62.5%        | 498 s         |
| VGG16 (pretrained)   | 86.5%        | 300 s         |
| VGG16 (fine-tuned)   | 88.7% (peak) | 314 s         |



# OVERFITTING IN FINE-TUNED VGG16

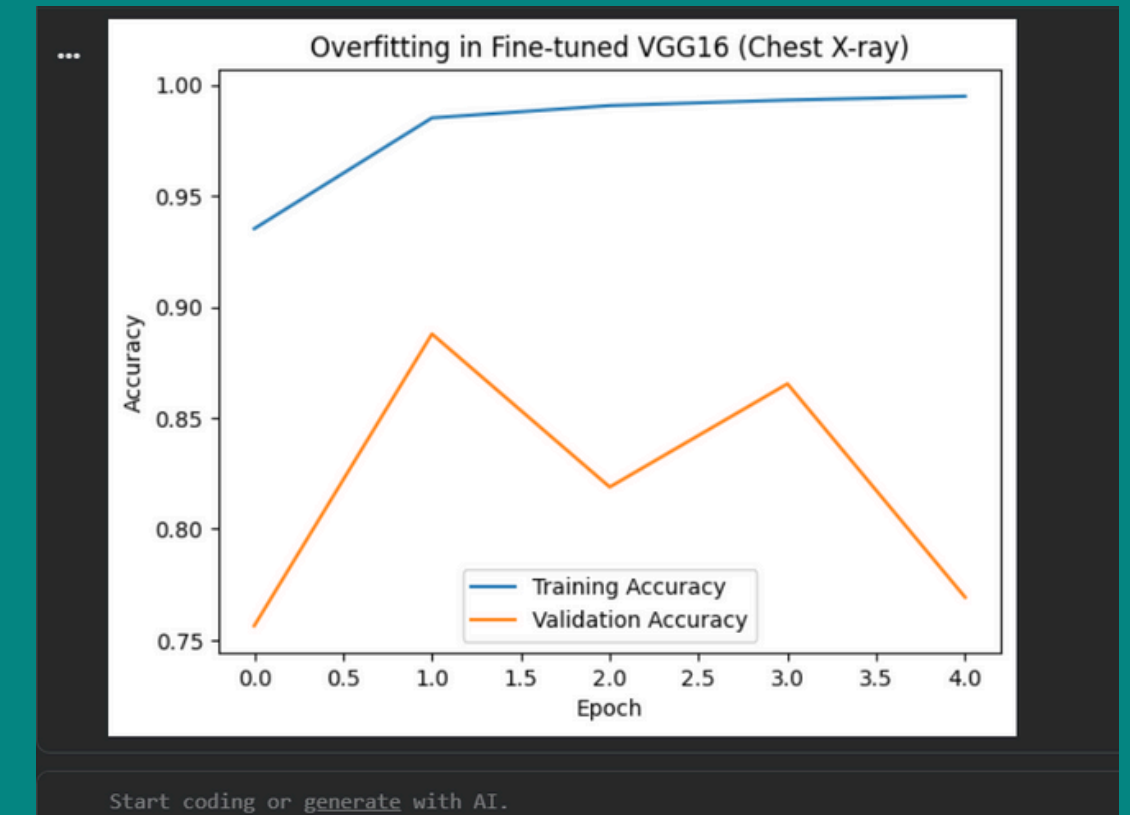
```
.. Epoch 1/5
164/164 ————— 69s 395ms/step - accuracy: 0.8658 - loss: 0.3010 - val_accuracy: 0.7564 - val_loss: 0.9452
Epoch 2/5
164/164 ————— 61s 371ms/step - accuracy: 0.9810 - loss: 0.0499 - val_accuracy: 0.8878 - val_loss: 0.4132
Epoch 3/5
164/164 ————— 62s 375ms/step - accuracy: 0.9915 - loss: 0.0252 - val_accuracy: 0.8189 - val_loss: 0.8094
Epoch 4/5
164/164 ————— 61s 369ms/step - accuracy: 0.9944 - loss: 0.0129 - val_accuracy: 0.8654 - val_loss: 0.7398
Epoch 5/5
164/164 ————— 61s 369ms/step - accuracy: 0.9956 - loss: 0.0129 - val_accuracy: 0.7692 - val_loss: 1.7470
Training Time: 313.69623613357544
```

Final epoch accuracy dropped to 76.9%

That means overfitting happened.

The model learned training data too well and lost generalization.

The fine-tuned VGG16 model shows increasing training accuracy but decreasing validation accuracy, indicating overfitting on the chest X-ray dataset.



Training accuracy keeps rising  
Validation accuracy goes up then drops → overfitting



# WHY PRETRAINED MODELS PERFORM BETTER

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Pretrained models have already learned low-level features such as edges, shapes, and textures from millions of images, which helps them adapt faster to new datasets like medical X-rays.



# CONCLUSION

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Pretrained VGG16 provides the best balance between accuracy and training efficiency for small datasets. Fine-tuning improves peak performance but requires careful regularization to avoid overfitting.

# CATS VS DOGS DATASET

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This is a non-medical natural image dataset used to evaluate the performance of CNN models on real-world object classification. It contains two classes: Cat and Dog. The dataset is more visually diverse and complex than medical images, making it suitable for testing generalization ability.



# PERFORMANCE ON CATS VS DOGS

| Model                | Validation Accuracy | Training Time |
|----------------------|---------------------|---------------|
| VGG16 (from scratch) | 50%                 | 1561 s        |
| VGG16 (pretrained)   | 93.1%               | ~700 s        |



# ANALYSIS OF CATS VS DOGS RESULTS

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The model trained from scratch performed no better than random guessing, indicating that deep CNNs require large datasets or pretrained weights. The pretrained VGG16 achieved very high accuracy, showing that transfer learning is extremely effective for natural image classification.

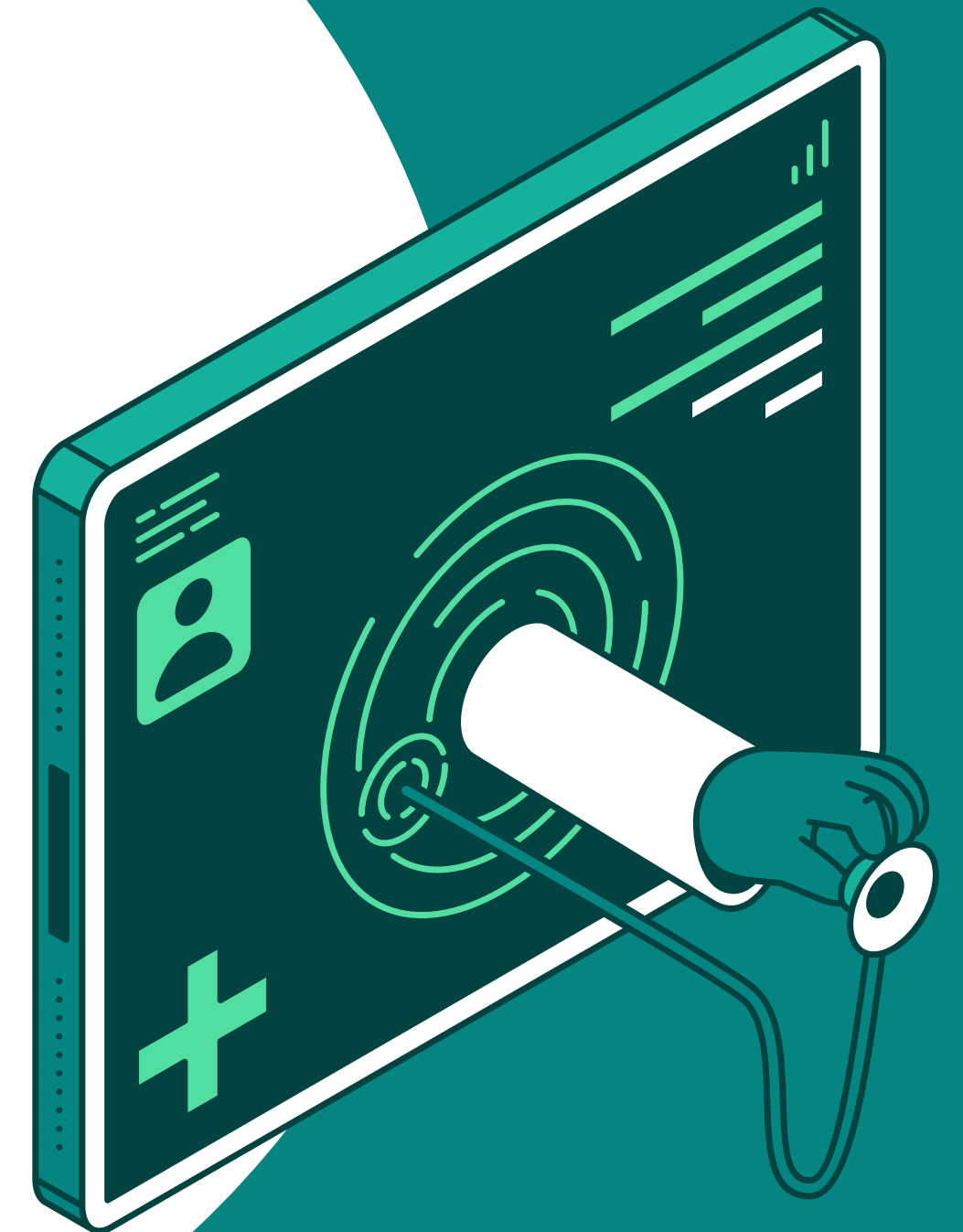




# MEDICAL VS NON-MEDICAL COMPARISON

| Dataset      | Best Model       | Accuracy |
|--------------|------------------|----------|
| Chest X-ray  | Pretrained VGG16 | ~86–88%  |
| Cats vs Dogs | Pretrained VGG16 | ~93%     |

The CNN achieved higher accuracy on natural images compared to medical images, indicating that medical image classification is more challenging due to subtle visual patterns.



# FINAL CONCLUSION

Pretrained VGG16 provides the best balance between accuracy, training time, and stability for both medical and non-medical image classification. Fine-tuning can improve peak performance, but it increases the risk of overfitting, especially for small medical datasets. Training deep CNNs from scratch is not suitable when the dataset size is limited. This study confirms that transfer learning is essential for achieving reliable and efficient performance in practical deep learning applications.

Code-

[https://drive.google.com/file/d/1eUwJ07Cwm2h4dMlBFdLRu7QtUYyykTfh/view?usp=drive\\_link](https://drive.google.com/file/d/1eUwJ07Cwm2h4dMlBFdLRu7QtUYyykTfh/view?usp=drive_link)

